Independent method checks and reproduction limits for the Campi Flegrei earthquake catalog

Source-verified computational assessment

9 October 2026

Abstract

Tan et al. develop an observational earthquake catalog and analyze its space–time distribution, magnitudes, focal mechanisms, and shallow hybrid events. The paper does not prescribe a coupled physical forward model. Both supplied PDFs are readable, including the supplementary methods and all figure captions. The uploaded archives contain derived catalogs, picks, focal solutions, and an interactive plot; the project restrictions exclude these as analytical inputs. I independently implement the documented relative-amplitude, signal-to-noise, and frequency-index equations and test a stated least-squares magnitude convention. Analytic consistency tests pass, but no empirical manuscript or supplementary panel is reproduced. This report identifies every figure and caption-defined panel or view, records its input blocker, and distinguishes generated method checks from observational reproduction.

1 Access, identity, and admissible inputs

The primary sources are the local files `tanmodel.pdf` and `tanmodelsupp.pdf`, supplied on the repository's main branch. The main document contains seven PDF pages, including the publication end page; the matching supplement contains 35 pages. Their title, authors, publication identifier, and supplementary contents agree. Main-text method descriptions and captions for Figures 1–4 were read, as were supplementary pages S2–S11, captions S1–S15, Tables S1–S2, and the archive descriptions on S28. Two font-weight warnings from the main PDF do not prevent text extraction; the supplementary text extracts without warnings.

File	Bytes	Inspection
`tanmodel.pdf`	2140554	Text readable; 7 pages
`tanmodelsupp.pdf`	62164947	Text readable; 35 pages
`science.adw9038_data_s1_to_s3.zip`	12960255	Directory readable; CRC passed; not extracted or executed
`science.adw9038_interactive_plot_s1.zip`	8852485	Directory readable; CRC passed; not extracted or executed

The first ZIP lists three CSV files, `science.adw9038_data_s1.csv`, `science.adw9038_data_s2.csv`, and `science.adw9038_data_s3.csv`, together with macOS resource-fork metadata. The supplement identifies these as the earthquake catalog, associated phases, and focal-mechanism solutions. The second ZIP contains `interactive_plot_s1.html` and macOS metadata; the supplement describes it as a three-dimensional seismicity visualization. Both ZIPs pass CRC integrity checks. Neither archive was

extracted; no archive entry was executed, and no numerical CSV rows or HTML code were consulted as analytical input.

The supplied archives do not remove the prohibition on original source data or derived products. The numerical workflow reads only independently chosen analytic fixtures. All four uploads are preserved unchanged, with exact SHA-256 values in `docs/file_inventory.json`. Published spatial figures were used as reference views to check correspondence between captions and panel layouts, including the inconsistent S7 letter references. No plot values were digitized, no published image is included here, and no published figure file enters a calculation.

2 Scientific premise and documented workflow

The observational target is the 21 January 2022–20 March 2025 seismic sequence. The study uses 23 broadband stations, nine short-period stations, and 16 accelerometers (supplement S2). A retrained PhaseNet picker supplies phase probabilities; GaMMA associates arrivals; VELEST and TomoDD refine locations. Subsequent analyses estimate magnitudes, focal mechanisms, and hybrid-event properties. Discussions of pressure, fluid migration, and ring-fault behavior are interpretations of these observations, not a system of coupled evolution equations with specified forcing and boundary conditions.

The final retraining pools contain 24,547 local earthquake traces, 145,292 noise traces, and 102,620 STEAD traces. The methods specify 30-s windows, up to five overlapping events, mean-squared-error loss, 60 epochs, and learning rate 0.001. Detection uses a probability threshold of 0.3 and excludes S picks at CASE, CFB1, and CFB3. GaMMA requires at least six picks and yields 105,944 candidate events. These are documentary counts; none is recreated here.

VELEST calibration uses 373 selected events and two runs that refine the one-dimensional velocity model, locations, and then station corrections. Three further fixed-model runs relocate the catalog, with residual cutoffs of 0.4 s and 0.3 s. Final acceptance requires at least six picks, RMS at most 0.25 s, and azimuthal gap at most 300 degrees. The reported intermediate catalog has 61,809 events. Two TomoDD rounds use a fixed three-dimensional velocity model and both picked and cross-correlated differential times, yielding 54,319 events. The routine catalog has 12,083 events. Their count ratio is approximately 4.50, an arithmetic comparison of published totals rather than a measured detection gain from this implementation.

Table S1 supplies a layered velocity parameter model. However, the three-dimensional field, actual picks, waveforms, and selected event subsets needed for the relocation results are excluded original inputs. Retraining without the study's admissible observations would not recreate its network weights or event detections. The complete detection, association, and relocation workflow is therefore not implemented or claimed as validated. The implementation below is limited to explicitly documented analysis equations and their consistency tests.

3 Independent implementation

3.1 Amplitude, power, and magnitude

Supplementary equations S2 and S5 define a relative magnitude from positive amplitude ratios:

$$d_{ij} = \log_{10}(A_i/A_j). \tag{1}$$

Comparable propagation and station response are required for this difference to represent a magnitude difference. Equal amplitudes give zero; a common multiplicative station gain cancels. The absolute magnitude cannot be determined from these differences alone. Equation S1 includes attenuation and

station corrections; its full coefficients and absolute calibration are not supplied as an independently executable specification here.

Equation S3 defines the signal-to-noise ratio as

$$\mathrm{SNR} = 10\log_{10}(P_{\mathrm{signal}}/P_{\mathrm{noise}}). \tag{2}$$

The written windows are 5 s after and 2 s before the actual or theoretical P arrival, with a 5 dB cutoff. I use mean-square power in each window. Integrated energy would introduce a spurious window-length ratio. The consistency test uses exact sinusoids with a factor-two amplitude difference, predicting $10\log_{10} 4$ dB despite the unequal window lengths. No arrival time is inferred from a measured trace.

The magnitude methods describe a weighted least-squares inversion with duration-magnitude regularization, but not its complete objective or regularization strength. I test the explicit convention

$$J(\mathbf{m}) = \sum_{i<j} w_{ij}(m_i - m_j - d_{ij})^2 + \lambda \sum_i (m_i - p_i)^2, \tag{3}$$

where p_i are assumed prior magnitudes. This is a documented implementation choice, not a claim about the exact authors' objective. The small graph-Laplacian normal system is solved by Gaussian elimination with partial pivoting in C++. The authors describe gradient descent; the exact linear solve here tests a specified quadratic problem instead.

Four analytic test magnitudes are $[0, 0.3, -0.2, 0.8]$, with amplitudes 10^{m_i} and weights $w_{ij} = 1/(j - i)$. With matching priors and $\lambda = 1$, the solution should recover those magnitudes. A common prior shift must shift the solution equally. With $\lambda = 0$, the global offset is undetermined, and the solver rejects the singular problem. A separate sensitivity calculation adds prior errors $[0.2, -0.1, 0.1, -0.2]$ and varies λ from 0.001 to 100. These arbitrary fixture values are not inferred from the publication and do not estimate its magnitude errors.

3.2 Frequency index

Supplementary equation S4 defines

$$\mathrm{FI} = \log_{10}(P_{8\text{–}15\,\mathrm{Hz}}/P_{2\text{–}5\,\mathrm{Hz}}). \tag{4}$$

The paper specifies an 8-s CAAM waveform window and the two bands. Sampling, taper, spectral normalization, and endpoint conventions are not fully stated. For the consistency tests I choose 100 Hz sampling, a rectangular window, mean removal, and inclusive frequency-bin boundaries. A direct double-precision discrete Fourier transform gives one-sided powers, normalized by N^2 and doubled except at DC and Nyquist. The common normalization cancels in FI; Parseval's identity provides a separate power check.

For the analytic signal

$$x(t) = \sin(2\pi\, 3t) + r\sin(2\pi\, 10t), \quad 0 \le t < 8\ \mathrm{s}, \tag{5}$$

the two tones fall on exact Fourier bins, so $\mathrm{FI} = 2\log_{10} r$. Their frequencies are test assumptions within the documented bands. The calculation does not generate an earthquake waveform. Common amplitude gain and circular time shifts must preserve FI. The 1–10 Hz display filter described for S14 is not imposed on the signal used to estimate the separate 8–15 Hz FI band.

There are no physical initial or boundary conditions: these are finite-window signal operations and a finite-dimensional inverse problem. No magma source, pressure history, fault law, or synthetic spatial earthquake distribution is imposed.

4 Validation and independently generated panels

All 27 checks passed. The maximum absolute FI error across the analytic mixtures was 4e-15; the maximum exact-recovery magnitude error was 1.94e-16. These errors compare against analytic fixtures, not observed earthquakes. The tests include exact ratios, ratio reversal, common-gain cancellation, invalid-amplitude rejection, spectral orthogonality, Parseval power conservation, FI shift/gain invariance, unequal-window SNR, pair-order invariance, and magnitude-offset ambiguity. Analytic tests constrain the implemented numerical operations; they do not validate performance on low-SNR field data, training, event association, relocation uncertainties, or a volcanic interpretation.

For each M1 panel, the data command is

```
./build/method_checks config/method_checks.cfg data
```

and the plotting command is

```
python3 scripts/plot_methods.py
```

M1a and M1b read `data/scaling.csv`; M1c reads `data/frequency_index.csv`; M1d reads `data/regularization.csv`. The analytic test waveform is additionally saved as `data/waveform.csv`. These panels are counted as four independently generated method checks and zero reproduced publication panels.

5 Discrepancies in the written specification

The uncertainty discussion on S5 refers to S7 panel letters H and I, whereas the figure and its caption use A–G: B/D/F give pick count versus error, and C/E/G give error densities. The panel record follows the figure and caption. Figure S9's methods retain events with NCCP+NCCS equal to 100, whereas its caption states a strict greater-than condition. Both predicates are recorded at 99, 100, and 101 in `data/thresholds.csv`; neither is applied to the prohibited catalog. The interactive plot has a different documented threshold, NCCP+NCCS+NCTP+NCTS greater than 1000, which must not be substituted for the S9 condition.

The focal-mechanism methods discuss 74 events, while the Data S3 caption describes 55 solutions. The archive was not mined to reconcile that difference. S15 uses the station spelling CCAM once, while the surrounding methods and normalization refer to CAAM. Remaining implementation gaps include exact magnitude-regularization strength, the cross-correlation weight convention, complete training splits/seeds and augmentation rejection rules, relocation damping and iteration settings, geographic polygons and time bins, event-dependent inversion bands, and kernel-density bandwidths. These are separate from the stronger excluded-input blocker.

6 Comparison with the publication

No empirical manuscript or supplementary panel is reproduced. No panel is classified as partially reproduced merely because its equation has been tested on analytic input. The complete main and supplementary figure inventory appears below. All rows have status *not reproduced*; their data and plot commands are *unavailable*, because no permitted empirical data-generation run could be specified. The JSON record retains these fields explicitly as null and records each stopped input assessment.

Published targets include the 54,319-event total, 38,103 best-located events, a completeness magnitude of 0.4 and $b = 0.97$, northern ring depths of 2.1–2.5 km, deeper southwestern activity to 3.7 km, and 70 identified hybrid events. These are documentary scales, not this implementation's outputs. The

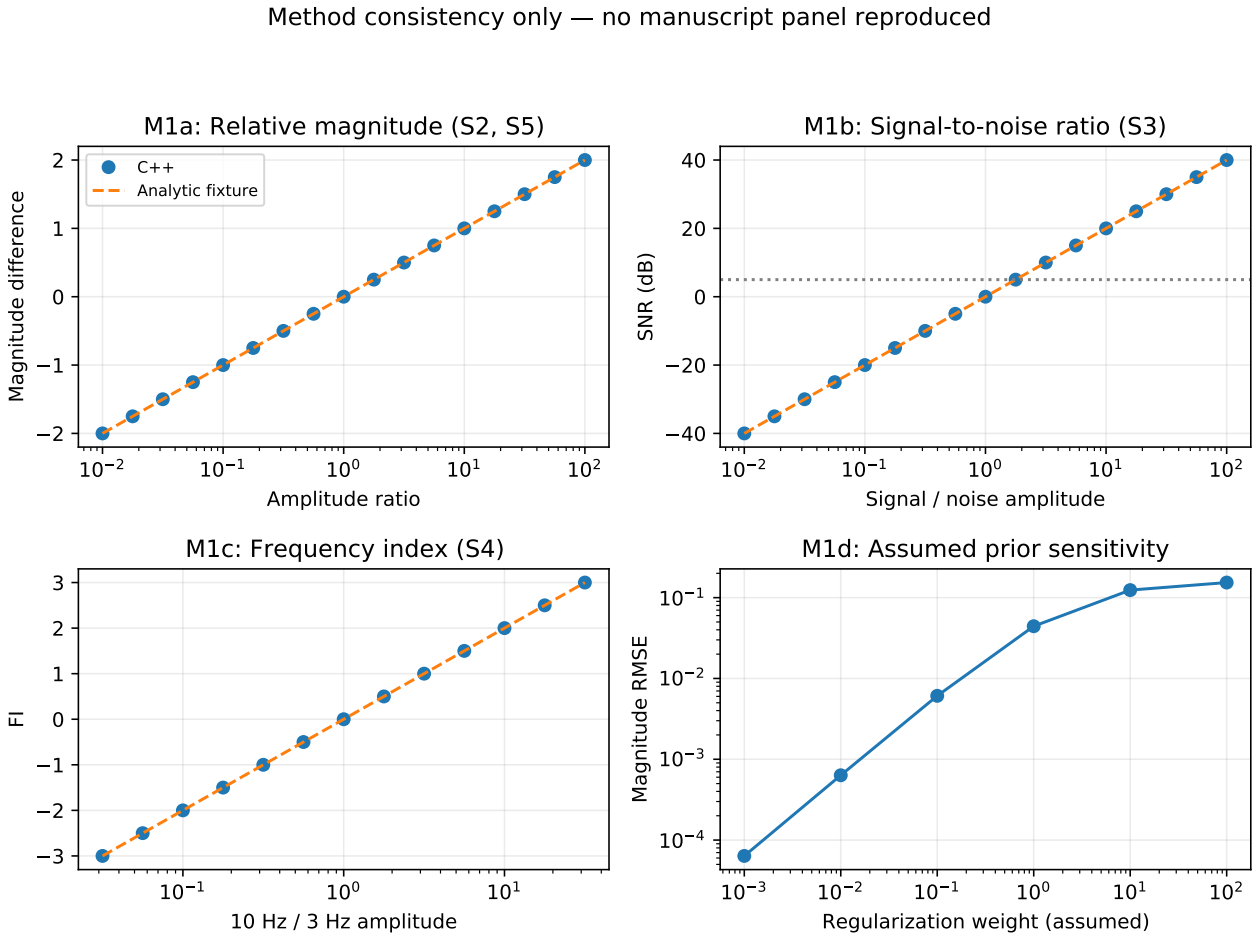


Figure 1: Independent method consistency panels, not manuscript reproductions. M1a: computed relative magnitude versus amplitude ratio, with the analytic ratio prediction. M1b: computed SNR for amplitude ratios; the dotted line marks the written 5 dB cutoff. M1c: FI calculated from 8-s mixtures of 3 Hz and 10 Hz tones, compared with $2\log_{10} r$. M1d: magnitude RMSE versus the assumed regularization strength when the four priors are biased. All points are plotted from saved C++ outputs. No event, trace, or value from either ZIP enters these panels.

source specification also distinguishes a deeper 4.8-km cluster outside the ring; therefore the 3.7-km conclusion must not be presented as a universal depth cutoff for all surrounding seismicity.

Catalog-count error, temporal alignment, spatial overlap, ring thickness, event-wise location error, focal-mechanism agreement, and hybrid-waveform similarity cannot be computed without permitted matched observations or results. Their absence is not zero error. FI errors and magnitude errors reported here are errors against analytic fixtures only. The generated figures do not test the reported FI range of -3.32 to 1.09 or the paper's inferred temporal evolution.

7 Rebuild, replacement, and remaining work

Run `python3 scripts/rebuild.py -clean` from the repository root. Before removing any generated products, it verifies the checksums of all four uploads and confirms PDF text/caption readability. It then removes only named, generated products; compiles C++; runs the checks; saves the numerical data; plots those data; builds the panel and validation tables; and compiles this report twice. It checks numerical assertions, PDF content, and layout warnings. Actual commands and exit codes are saved in `data/build_log.json`. Tool versions, verification results, and the separate scientific-incompletion status are in `data/build_status.json`. File hashes are in `data/manifest.sha256`. No network access is required.

The previous travel-time C++ example, its configuration, validation figures, numerical CSVs, plotting/audit scripts, and report tables were removed. The interrupted access-only report was superseded. New source, configuration, CSVs, method figures, panel inventory, documentation, LaTeX, and PDF replace those results. The exact replacement list is `docs/REPLACEMENTS.md`. The user-uploaded PDFs and ZIPs, permitted guidance files, repository identity, and all Git history remain intact. This is a corrected evidence-based assessment with validated method components, not a completed empirical reproduction.

All project intermediates and caches remain within the repository: `tmp/`, `build/`, `data/`, `figures/`, and `report/`. Existing system Python/compiler/Poppler installations and TeX resources under `/usr/share/texlive` are used. No dependency installation or external project cache is created. Further empirical progress requires inputs compatible with the stated restrictions; the supplied original catalogs and interactive plot do not meet that condition.

A Numerical checks

Errors and tolerances are dimensionless for logarithmic quantities and magnitude units for magnitude recovery; Parseval compares normalized signal power. Zero-tolerance checks record whether invalid or singular inputs were correctly rejected. These tolerances are implementation acceptance criteria, not observational uncertainties.

Check	Absolute error	Tolerance
equal_amplitudes	0	1e-14
decade_amplitude	0	1e-14
ratio_reversal	0	1e-14
common_station_gain	5.55e-17	1e-14
zero_amplitude_rejected	0	0
negative_power_rejected	0	0
fi_mixture_0	4e-15	1e-10
fi_mixture_1	8.88e-16	1e-10
fi_mixture_2	1.78e-15	1e-10
fi_mixture_3	1.55e-15	1e-10
fi_mixture_4	1.78e-15	1e-10
fi_mixture_5	2.11e-15	1e-10
fi_mixture_6	2.22e-16	1e-10
fi_mixture_7	2.11e-15	1e-10
fi_mixture_8	1.33e-15	1e-10
fi_mixture_9	8.88e-16	1e-10
fi_mixture_10	1.33e-15	1e-10
fi_mixture_11	4.44e-16	1e-10
fi_mixture_12	1.78e-15	1e-10
parseval	3.33e-15	1e-10
fi_circular_shift	7.22e-16	1e-10
fi_gain	8.88e-16	1e-10
snr_5s_2s_windows	8.88e-16	1e-10
magnitude_exact_recovery	1.94e-16	1e-12
magnitude_pair_order	0	1e-12
magnitude_gauge_shift	4.44e-16	1e-12
unanchored_problem_rejected	0	0

B Figure-by-figure record

Every main/supplementary figure and its caption-defined panels are represented, including unlettered spatial views. Waveform/spectrogram components are covered within their labeled example. S10 rows are identified by year, S11 by positional swarm number, and S14 by cluster because those descriptors are explicit in the captions. For *every row*, status is *not reproduced*, data command is *unavailable*, and plot command is *unavailable*. Only the M1 method-check panels have executable generation commands, given above. This distinction prevents a metadata audit from being counted as a numerical reproduction attempt.

Panel/view	Quantity	Specification, parameters and blocker
1A	INGV event map and large-event markers	Main p.71, Fig.1 caption. 2005-01-01 to 2025-03-20; origin 14.14 E, 40.82 N; sea-level depth; Md>=4 markers. Blocked: Excluded observed or author-derived earthquake catalog.
1B	Vertical displacement at Pozzuoli port	Main p.71, Fig.1 caption. 1905-2023; Rione Terra. Blocked: Excluded observed displacement series from cited monitoring sources.
1C	Temporal seismicity and cumulative counts	Main p.71, Fig.1 caption. 2005-2025; cumulative Md>0.2; study interval 2022-01-21 to 2025-03-20. Blocked: Excluded observed or author-derived earthquake catalog.
2A	Catalog seismicity rates by region	Main p.72, Fig.2 caption. Central vs ring/other regions; 54319 vs 12083 catalog events; time bin unspecified. Blocked: Excluded observed or author-derived earthquake catalog. Exact regional polygons and time bin not tabulated.
2B	Depth versus time, symbol size by magnitude	Main p.72, Fig.2 caption. Sea-level depth; near-surface 0-0.5 km; central 0.5-2.1; ring sectors to 3.7 km. Blocked: Excluded observed or author-derived earthquake catalog. Hybrid-event dates and locations are observed outputs.
3/map	Epicentral spatial pattern	Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. Blocked: Excluded observed or author-derived earthquake catalog. Note: Views are not lettered in the caption.
3/northing-depth	Depth versus northing	Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. Blocked: Excluded observed or author-derived earthquake catalog. Note: Views are not lettered in the caption.
3/easting-depth	Depth versus easting	Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. Blocked: Excluded observed or author-derived earthquake catalog. Note: Views are not lettered in the caption.
4A	Ring-sector focal mechanisms and seismicity	Main p.74, Fig.4; supplement S6-S7. High quality: SKHASH A or Kagan angle<30 deg; gCAP preference with stated exception. Blocked: Excluded polarities, amplitude ratios, waveforms, and focal solutions.
4B	Central/onshore focal mechanisms and seismicity	Main p.74, Fig.4; supplement S6-S7. High quality: SKHASH A or Kagan angle<30 deg; gCAP preference with stated exception. Blocked: Excluded polarities, amplitude ratios, waveforms, and focal solutions.
S1	Station data-availability timeline	Supplement S12 caption; S2 methods. 48 stations; continuous-data availability. Blocked: Excluded original waveforms/picks and station metadata.
S2	Station map	Supplement S13 caption; S2 methods. 23 broadband, 9 short-period, 16 accelerometers. Blocked: Excluded original waveforms/picks and station metadata.
S3A	Noise example: waveform, phase labels, spectrogram	Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Blocked: Excluded original waveforms/picks and station metadata. Training examples cannot be recreated without allowed waveform inputs.
S3B	Single-event example: waveform, phase labels, spectrogram	Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Blocked: Excluded original waveforms/picks and station metadata. Training examples cannot be recreated without allowed waveform inputs.

Panel/view	Quantity	Specification, parameters and blocker
S3C	Two overlapping events: waveform, phase labels, spectrogram	Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Blocked: Excluded original waveforms/picks and station metadata. Training examples cannot be recreated without allowed waveform inputs.
S3D	Five overlapping copies of one event: waveform, phase labels, spectrogram	Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Blocked: Excluded original waveforms/picks and station metadata. Training examples cannot be recreated without allowed waveform inputs.
S4A	Swarms at CSOB: waveform, spectrogram, original and retrained picker scores	Supplement S15 caption; S2-S3. 120 s; ~2 h after May 2024 Md4.4 event; phase threshold 0.3. Blocked: Excluded original waveforms/picks and station metadata. Original/retrained weights and outputs excluded.
S4B	CMIS noise: waveform, spectrogram, original and retrained picker scores	Supplement S15 caption; S2-S3. 1600 s; coastal noise; phase threshold 0.3. Blocked: Excluded original waveforms/picks and station metadata. Original/retrained weights and outputs excluded.
S5/map	Routine-catalog spatial views; map	Supplement S16 caption. 12083 events; 1D model and HYPO71. Blocked: Excluded observed or author-derived earthquake catalog. Note: Unlettered spatial view. Published reference used only to check correspondence with the written caption; no numerical values digitized.
S5/northing-depth	Routine-catalog spatial views; northing-depth	Supplement S16 caption. 12083 events; 1D model and HYPO71. Blocked: Excluded observed or author-derived earthquake catalog. Note: Unlettered spatial view. Published reference used only to check correspondence with the written caption; no numerical values digitized.
S5/easting-depth	Routine-catalog spatial views; easting-depth	Supplement S16 caption. 12083 events; 1D model and HYPO71. Blocked: Excluded observed or author-derived earthquake catalog. Note: Unlettered spatial view. Published reference used only to check correspondence with the written caption; no numerical values digitized.
S6/map	Initial-location spatial views; map	Supplement S17 caption. 61809 events; five VELEST runs. Blocked: Excluded observed or author-derived earthquake catalog. Note: Unlettered spatial view. Published reference used only to check correspondence with the written caption; no numerical values digitized.
S6/northing-depth	Initial-location spatial views; northing-depth	Supplement S17 caption. 61809 events; five VELEST runs. Blocked: Excluded observed or author-derived earthquake catalog. Note: Unlettered spatial view. Published reference used only to check correspondence with the written caption; no numerical values digitized.
S6/easting-depth	Initial-location spatial views; easting-depth	Supplement S17 caption. 61809 events; five VELEST runs. Blocked: Excluded observed or author-derived earthquake catalog. Note: Unlettered spatial view. Published reference used only to check correspondence with the written caption; no numerical values digitized.
S7A/map	Cluster locations and 2-sigma errors: map and cross-sectional views; map	Supplement S18 caption; S5 methods. Central174, north165, southwest175, east166; HypoDD SVD; Table S1. Blocked: Excluded original waveforms/picks and station metadata. Relocated subsets and error estimates excluded.
S7A/easting-depth	Cluster locations and 2-sigma errors: map and cross-sectional views; easting-depth	Supplement S18 caption; S5 methods. Central174, north165, southwest175, east166; HypoDD SVD; Table S1. Blocked: Excluded original waveforms/picks and station metadata. Relocated subsets and error estimates excluded.

Panel/view	Quantity	Specification, parameters and blocker
S7B	EX uncertainty versus differential-pick count	Supplement S18 caption. Log-log axes; NCCP+NCCS. Blocked: Excluded observed or author-derived earthquake catalog. HypoDD covariance outputs unavailable as permitted input. Note: Caption labeling retained; methods prose uses inconsistent letter references.
S7D	EY uncertainty versus differential-pick count	Supplement S18 caption. Log-log axes; NCCP+NCCS. Blocked: Excluded observed or author-derived earthquake catalog. HypoDD covariance outputs unavailable as permitted input. Note: Caption labeling retained; methods prose uses inconsistent letter references.
S7F	EZ uncertainty versus differential-pick count	Supplement S18 caption. Log-log axes; NCCP+NCCS. Blocked: Excluded observed or author-derived earthquake catalog. HypoDD covariance outputs unavailable as permitted input. Note: Caption labeling retained; methods prose uses inconsistent letter references.
S7C	EX uncertainty density by cluster	Supplement S18 caption. Kernel density estimate; bandwidth unspecified. Blocked: Excluded observed or author-derived earthquake catalog. HypoDD covariance outputs excluded. Note: Caption labeling retained; methods prose also refers to panels H/I.
S7E	EY uncertainty density by cluster	Supplement S18 caption. Kernel density estimate; bandwidth unspecified. Blocked: Excluded observed or author-derived earthquake catalog. HypoDD covariance outputs excluded. Note: Caption labeling retained; methods prose also refers to panels H/I.
S7G	EZ uncertainty density by cluster	Supplement S18 caption. Kernel density estimate; bandwidth unspecified. Blocked: Excluded observed or author-derived earthquake catalog. HypoDD covariance outputs excluded. Note: Caption labeling retained; methods prose also refers to panels H/I.
S8A	Estimated local versus routine duration magnitude	Supplement S19 caption; S5-S6 methods. Relative-amplitude LS; SNR>=5 dB; >=2 common stations; separation<distance/5. Blocked: Excluded original waveforms/picks and station metadata. Both magnitude catalogs excluded; regularization strength unspecified.
S8B	Magnitude-frequency distribution and fitted b value	Supplement S19 caption; S6 methods. Published Mc=0.4 by maximum curvature; b=0.97 by ML; bin size unspecified. Blocked: Excluded observed or author-derived earthquake catalog.
S9/map	Filtered-catalog spatial views and feature labels A-D; map	Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. Blocked: Excluded observed or author-derived earthquake catalog. Note: Unlettered spatial view. Published reference used only to check correspondence with the written caption; no numerical values digitized.
S9/northing-depth	Filtered-catalog spatial views and feature labels A-D; northing-depth	Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. Blocked: Excluded observed or author-derived earthquake catalog. Note: Unlettered spatial view. Published reference used only to check correspondence with the written caption; no numerical values digitized.

Panel/view	Quantity	Specification, parameters and blocker
S9/easting-depth	Filtered-catalog spatial views and feature labels A-D; easting-depth	Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. Blocked: Excluded observed or author-derived earthquake catalog. Note: Unlettered spatial view. Published reference used only to check correspondence with the written caption; no numerical values digitized.
S10/2022	Annual seismicity map	Supplement S21 caption. Year 2022. Blocked: Excluded observed or author-derived earthquake catalog.
S10/2023	Annual seismicity map	Supplement S21 caption. Year 2023. Blocked: Excluded observed or author-derived earthquake catalog.
S10/2024	Annual seismicity map	Supplement S21 caption. Year 2024. Blocked: Excluded observed or author-derived earthquake catalog.
S10/2025	Annual seismicity map	Supplement S21 caption. Year 2025 through March20. Blocked: Excluded observed or author-derived earthquake catalog.
S11/swarm-1	Map of one of four major swarms	Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Blocked: Excluded observed or author-derived earthquake catalog. Note: Positional identifiers for four maps; event windows not inferred from plotted data.
S11/swarm-2	Map of one of four major swarms	Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Blocked: Excluded observed or author-derived earthquake catalog. Note: Positional identifiers for four maps; event windows not inferred from plotted data.
S11/swarm-3	Map of one of four major swarms	Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Blocked: Excluded observed or author-derived earthquake catalog. Note: Positional identifiers for four maps; event windows not inferred from plotted data.
S11/swarm-4	Map of one of four major swarms	Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. Blocked: Excluded observed or author-derived earthquake catalog. Note: Positional identifiers for four maps; event windows not inferred from plotted data.
S12A	Ring-sector division	Supplement S23 caption; S8. East, southwest, north; sector boundaries by location/depth. Blocked: Excluded observed or author-derived earthquake catalog. Exact boundary coordinates not tabulated.
S12B	Ring-sector division in complementary spatial view	Supplement S23 caption; S8. East, southwest, north; sector boundaries by location/depth. Blocked: Excluded observed or author-derived earthquake catalog. Exact boundary coordinates not tabulated.
S12C	Common-receiver gather: east / HHZ	Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. Blocked: Excluded original waveforms/picks and station metadata.
S12D	Common-receiver gather: southwest / HHZ	Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. Blocked: Excluded original waveforms/picks and station metadata.
S12E	Common-receiver gather: north / HHZ	Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. Blocked: Excluded original waveforms/picks and station metadata.

Panel/view	Quantity	Specification, parameters and blocker
S12F	Common-receiver gather: east / HHN	Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. Blocked: Excluded original waveforms/picks and station metadata.
S12G	Common-receiver gather: southwest / HHN	Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. Blocked: Excluded original waveforms/picks and station metadata.
S12H	Common-receiver gather: north / HHN	Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. Blocked: Excluded original waveforms/picks and station metadata.
S13A	Full caldera seismicity on satellite reference	Supplement S24 caption. Depth-coded relocated events; Google Earth basemap. Blocked: Excluded observed or author-derived earthquake catalog. Published imagery not reused.
S13B	Onshore seismicity on satellite reference	Supplement S24 caption. Depth-coded relocated events; Google Earth basemap. Blocked: Excluded observed or author-derived earthquake catalog. Published imagery not reused.
S14/1	CAAM waveforms and event annotations	Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. Blocked: Excluded original waveforms/picks and station metadata. Event identities, correlations, FI and magnitudes are author-derived. Note: Cluster identifiers; not invented panel letters. No magnitude/correlation for unclustered events.
S14/2	CAAM waveforms and event annotations	Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. Blocked: Excluded original waveforms/picks and station metadata. Event identities, correlations, FI and magnitudes are author-derived. Note: Cluster identifiers; not invented panel letters. No magnitude/correlation for unclustered events.
S14/3	CAAM waveforms and event annotations	Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. Blocked: Excluded original waveforms/picks and station metadata. Event identities, correlations, FI and magnitudes are author-derived. Note: Cluster identifiers; not invented panel letters. No magnitude/correlation for unclustered events.
S14/unclustered	CAAM waveforms and event annotations	Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. Blocked: Excluded original waveforms/picks and station metadata. Event identities, correlations, FI and magnitudes are author-derived. Note: Cluster identifiers; not invented panel letters. No magnitude/correlation for unclustered events.
S15A	Hybrid family 1: waveforms and spectrograms	Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Blocked: Excluded original waveforms/picks and station metadata. Note: Caption spells CCAM once; interpreted as CAAM consistently with methods. Each entry covers the traces/spectrograms of that labeled example.

Panel/view	Quantity	Specification, parameters and blocker
S15B	Hybrid family 2: waveforms and spectrograms	Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Blocked: Excluded original waveforms/picks and station metadata. Note: Caption spells CCAM once; interpreted as CAAM consistently with methods. Each entry covers the traces/spectrograms of that labeled example.
S15C	Hybrid family 3: waveforms and spectrograms	Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Blocked: Excluded original waveforms/picks and station metadata. Note: Caption spells CCAM once; interpreted as CAAM consistently with methods. Each entry covers the traces/spectrograms of that labeled example.
S15D	Shallow volcano-tectonic example: waveforms and spectrograms	Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Blocked: Excluded original waveforms/picks and station metadata. Note: Caption spells CCAM once; interpreted as CAAM consistently with methods. Each entry covers the traces/spectrograms of that labeled example.

C Written sources

Tan, X., Tramelli, A., Gammaldi, S., Beroza, G. C., Ellsworth, W. L., and Marzocchi, W. (2025). *A clearer view of the current phase of unrest at Campi Flegrei caldera.* Science 390, 70–75. Local source: `tanmodel.pdf`. Matching supplementary methods, equations, captions, and tables: `tanmodelsupp.pdf`. The archive descriptions on supplementary page S28 identify the prohibited derived products. No original author implementation was consulted. The preserved `AGENTS.md` and academic style guide govern the minimal implementation and restrained reporting.